

The Primary Parental Investment in Children in the Contemporary USA is Education

Testing the Trivers-Willard Hypothesis of Parental Investment

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Abstract This paper tests the Trivers-Willard hypothesis that high-status individuals will invest more in sons and low-status individuals will invest more in daughters using data from the 2000 to 2010 General Social Survey and the 1979 National Longitudinal Survey of Youth. We argue that the primary investment U.S. parents make in their children is in their children's education, and this investment is facilitated by a diverse market of educational choices at every educational level. We examine two measures of this investment: children's years of education and the highest degree attained. Results show that sons of high-status fathers receive more years of education and higher degrees than daughters, whereas daughters of low-status fathers receive more years of education and higher degrees than sons. Further analyses of possible mechanisms for these findings yield null results. We also find that males are more likely to have high-status fathers than females.

Keywords Trivers-Willard · Parental investment · Education · Biosocial

The Trivers-Willard hypothesis predicts that individuals in good condition will invest more in male offspring, and individuals in poor condition will invest more in female offspring (Trivers 1972; Trivers and Willard 1973). Investment in offspring may be biological and include the energy and nutrients required for conception and gestation, most of which will come from the mother. Differential biological investment by sex will be manifest in sex ratios at birth. Investment in offspring may also be nonbiological and include the energy, time, and resources involved in the rearing of offspring, and this type of investment can come from either parent.

The hypothesis was designed to be applicable to all sexually reproducing species, but its authors specifically note that “the model can be applied to humans differentiated on a socioeconomic scale, as long as the reproductive success of a male at the upper end of the scale exceeds his sister's, while that of a female at the lower end of the scale

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exceeds her brother's" (Trivers and Willard 1973:91). Applied to humans the Trivers-Willard hypothesis thus relies on the sex difference in reproductive success at the top and bottom of the socioeconomic scale, of which there is substantial evidence in preindustrial societies (Betzig 2012). It has been suggested that this sex difference in reproductive success no longer exists in contemporary industrialized societies such that high-status men no longer out-reproduce high-status women (Keller et al. 2001). However, recent evidence suggests that high economic status does promote reproductive success for men but not for women (Fieder and Huber 2007; Hopcroft 2005; Nettle and Pollet 2008). In the U.S., lower-income men are less likely than higher-income men to marry (Oppenheimer 2000) and have children (Hopcroft 2005). In Europe, lower-income men are less likely to have children (Barthold et al. 2012). According to the Trivers-Willard hypothesis, given the sex difference in reproductive outcomes, it will be in the reproductive interests of individuals to differentially invest in their progeny by sex—in good conditions, invest in males; in poor conditions, invest in females.

Previous tests of the Trivers-Willard hypothesis have yielded mixed results, particularly for populations in developed societies (see Cronk 2007 for a review of studies that focus on nonbiological investment; Lazarus 2002 for a review of studies that focus on the sex ratio or biological investment). Results differ by whether the study tests the sex ratio version or nonbiological investment version of the hypothesis, the methodology and measures used, and the study population.

Studies in the U.S. not supporting the Trivers-Willard hypothesis of nonbiological investment include Freese and Powell (1999), who found no sex difference in investments in adolescents in the form of measures such as saving money for college, enrolling children in private school, having educational objects in the home, talking to children about school, or monitoring of children's behavior. Keller et al. (2001) also found no sex difference in investments in babies and small children in terms of hours per week participated with children, self-reported warmth, and months breast-fed. Beaulieu and Bugental (2008) found no support for the Trivers-Willard hypothesis from their analysis of early interactions between mothers and children. Schnettler (2011), using self-report data from the National Longitudinal Study of Adolescent Health, found support for the Trivers-Willard hypothesis only for certain types of parental investment and under specific circumstances. He found support for the hypothesis for reports of breastfeeding initiation but not continuation and for children's perceptions of how much their parents care for them. Schnettler (2011) found no support for the Trivers-Willard hypothesis for parental engagement in activities with their children (see also Nettle 2010).

These studies examine self-reports of activities with children or adolescents and/or investments in children or adolescents. Schnettler (2011) notes that such reports are subject to social desirability biases, but he also uses them for reports of breastfeeding initiation (but not for children's perceptions of how much their parents care for them). Yet not only are social desirability issues likely to be a problem with such reports in developed societies such as the U.S., but simple recall is likely to be a problem when parents are asked about how much time they spend doing things with their children.

Second, many of the studies of nonbiological parental investments focus on investments in children at a relatively young age (as infants, children, and adolescents). Yet measures of investments in children as adolescents or younger do not capture the entirety (or even the majority) of parental investment in children. Further, while

parental investment in young children is crucial for their survival, parental investments later in the child's life (such as in a college education) are more important for the child's adult success in a postindustrial society such as the U.S. (Hout 2012).

The Trivers-Willard hypothesis implies that the goal of any parental investment is to have children be successful so they can attract mates and provide the parents with high-quality grandchildren. This is an ends-oriented goal, not a means-oriented goal. That is, parents are likely to do whatever they can to help their children be successful, given the situation and their own resources. The Trivers-Willard hypothesis suggests that high-status parents will make an extra effort for boys, and low-status parents will make an extra effort for girls.

We argue that in contemporary American society the primary investment most parents make in their children is in their children's education. In the U.S., education is strongly positively correlated with economic success, health, family stability, and social connections (Hout 2012), so by investing in their children's education parents can help ensure their children's economic, social, and reproductive success. A study of the 2007–2009 earnings of men and women in their prime earning years (30–54) found that men's and women's annual earnings rose roughly 20% for each educational level (high school diploma, bachelor's degree, etc.) they had earned (Hout 2012). In a 40-year work life, white men with college degrees can expect to earn at least a million dollars more than their high school graduate counterparts (Julian and Kominski 2011). Education's role in boosting income is particularly important for men because, as noted above, income is particularly important for men's marital attainment and hence reproductive success. Education's role in boosting income is less important for women because their income tends to be negatively associated with number of offspring. However, education is important for women's reproductive success in other ways. There is abundant evidence of homogamy in the American marriage market whereby marriage partners tend to have similar amounts of education (Kalmijn 1998; Schwartz 2010), so an educated woman is more likely to have a highly educated, high-income-earning husband than her less-educated peer. This means that an investment in education is an investment in the marital prospects of all children, both male and female.

Cronk (2007) suggests that we should not focus on investments such as education in part because such investments "may involve nothing more special than a general ability to calculate costs and benefits." But predispositions such as those presumably involved in producing the Trivers-Willard effect are involved in all cost-benefit analyses because they shape preferences, and preferences help determine individual perceptions of both costs and benefits. So we think that investments in education cannot be excluded from inclusion in tests of the Trivers-Willard hypothesis.

This investment in a child's education, particularly if parents utilize private schools and colleges, can be expensive. Unlike in many other advanced industrial societies, the primary responsibility for the cost of higher education lies with the child and the child's family (although other societies have Americanized in this respect in recent years). Even though there are other sources of support for higher education (loans, scholarships, etc.) parental support is often instrumental in obtaining them (for example, being a cosigner on a loan, or searching out appropriate scholarships).

In the U.S., this investment is not misplaced (Hout 2012). A child's years of schooling and achievement of educational credentials are important avenues toward occupational success and hence overall lifetime success (although education is not a

guarantee of a job or success). Years ago, the sociologists Blau and Duncan (1967) showed that the link between father's occupational status and son's occupational status in the U.S. operated primarily through the son's years of schooling, a finding that has been replicated many times in the U.S. and other industrialized countries (Kerckhoff 1995).

In this paper, we test the Trivers-Willard hypothesis of nonbiological investment in children using amount of education received by the child as the measure of parental investment. This paper in part replicates Hopcroft (2005) using data from the years 2000 to 2010, it also includes additional analyses of highest credential attained. To further examine the mechanism by which higher-status parents help their sons and lower-status parents help their daughters attain more education, we include analyses of type of high school attended (public or private), tuition paid at first year of college, and total amount of college loans, using data from the 1979 National Longitudinal Survey of Youth. Unfortunately, we do not have information on the name of the college attended or whether it was public or private.

Another issue raised by previous studies is how best to measure parental status. Cronk (2007) notes that previous tests of the Trivers-Willard hypothesis have used "income, wealth, class or caste membership, ethnic status, educational attainment, husband-wife age differences, and occupational status." In the studies mentioned above Schnettler (2011) uses parental education; Keller et al. (2001) use family income, average educational levels of both parents, whether family had received food assistance, and presence of father; Freese and Powell (1999) use total family income and highest educational credential of the most highly educated parent.

We argue that many of the measures of parental status used in previous studies are not optimal measures. For example, income is a common measure of parental status. Yet some sources of income grant the earner (and the earner's family) greater social prestige than other sources of income, so income by itself it is not a good measure of status. Similarly education is not by itself an optimal measure of social status, because education by itself does not guarantee a high income or a prestigious job (although education is of course positively correlated with income and occupational prestige). Blau and Duncan (1967) solved this problem with their development of the socioeconomic index (SEI) for measuring the status of various occupations in the U.S. This measure is computed from measures of the average education, income, and social prestige associated with particular occupations (Blau and Duncan 1967), and sociologists periodically update the measure so as to reflect contemporary realities in the U.S. This is the measure we use.

Our use of the SEI measure is not entirely unproblematic. Given that it is a measure associated with the average characteristics of an occupation, it will not capture all dimensions of social status in the U.S. For example, it will not capture the status of people who have high social status but have no formal occupation of their own, notably women. Given that men are most likely to have full-time occupations, this measure is least problematic when applied to men, so Blau and Duncan (1967) used it for men only in their landmark work *The American Occupational Structure*, to examine how status was passed from father to son.

In this paper, we use father's SEI when the respondent was about 16 years old as the measure of familial status. This measure is determined by asking the respondent what their father did for a living when the respondent was about 16, and assigning a SEI

score to each occupation. The father may not be their biological father, but the SEI of a stepfather is still a measure of family status. In the contemporary U.S. it may seem problematic that we do not include a measure of mother's status. We argue that father's occupational status remains the best single measure of familial status. The majority of men are employed full-time, while this is less true of women (U.S. Bureau of Labor Statistics 2012), and employed men are more likely to be married (Oppenheimer 2000). Further, when both husband and wife are employed full-time, their occupational statuses tend to be similar (Kalmijn 1998).

We also test the Trivers-Willard hypothesis of biological investment in offspring using data from the 2000 to 2010 General Social Survey. This version of the hypothesis predicts that high-status individuals will have more sons and low-status individuals will have more daughters. Using those data we are not able to look at the biological children of respondents, but for each respondent we can determine what their father's status was when they were about 16 years old. Assuming that most respondents give the occupational status of their biological fathers (which is likely given the data), we can determine whether males are more likely to have high-status fathers than females by correlating respondent sex with father's status. If high-status fathers are more likely to have male children, and since this is a random sample of the U.S. population, there should be a positive correlation between sex (male=1) and the status of fathers when the respondent was about 16.

Data and Methods

The data are from the 2000 to 2010 General Social Surveys (GSS) and the 1979 National Longitudinal Survey of Youth (NLSY). The GSS is conducted at the National Opinion Research Center at the University of Chicago (Davis and Smith 1998). Surveys for the years 2000–2010 are combined for a total of 11,926 cases. Each survey is an independently drawn, multistage probability sample of noninstitutionalized, English-speaking persons aged 18 or over living in the United States. The GSS is a widely used survey of the American population that has been fielded every few years since 1972.

The NLSY is a longitudinal survey of a sample of American youth born between 1957 and 1964 fielded by the Bureau of Labor Statistics. The cross-sectional survey sample we use here was designed to represent the entire population born from 1957 to 1964. The sample originally included 6,111 respondents ages 14–22 when first interviewed in 1979 (note that we excluded the two oversamples in the NLSY79 from this analysis).

Dependent Variables

To measure parental investment in education, we first use respondent's *years of education* attained, which ranges from 0 to 20. This variable is available in both the GSS and NLSY.

With data from the NLSY, we also are able to determine whether the current or last junior or high school the child attended was a public or private school (*public*: public = 1, private = 0). We also examine *tuition79*, which is the full-time tuition and fees of the

current or most recent college the respondent was enrolled in since 1978. This sample size is small in the analysis of this variable because the NLSY only asked this question of respondents in 1979, at which time few of the respondents were in college. Also using the NLSY, we examine a loan variable, *loans1990*, which is the respondent's total amount of educational loans in 1990, at which time the respondents would have been ages 25–33.

For all of these dependent variables we use ordinary least squares regression analysis. This model is

$$\hat{Y} = a + b_1age + b_2race + b_3sex + b_4father's\ status + b_5siblings + b_6sex*father's\ status$$

The second measure of respondent education is respondent's *highest degree* obtained, which ranges from less than high school to graduate degree (<HS = 0, HS = 1, Junior College = 2, Bachelor's Degree = 3, Graduate/Professional Degree = 4). Since this is an ordinal dependent variable with discrete values we use an ordered-probit analysis. (Ordered logit models were inappropriate because the data did not meet the proportional odds assumption.). This model assumes an underlying continuum to the ordinal dependent variable and estimates threshold levels of this continuum, along with parameters for each of the independent variables. The following specification was used:

$$T_n^* = \beta'z_n + \varepsilon_n$$

where T_n^* is a latent and continuous measure of highest degree of respondent, z_n is a vector of explanatory variables describing the respondent, β is a vector of parameters to be estimated, and ε_n is a random error term.

The observed and coded highest degree, T_n , is determined from the model as follows.

$$\begin{aligned} T_n &= 0 \text{ if } -\infty \leq T_n^* \leq \mu_1 \text{ (< HS)} \\ T_n &= 1 \text{ if } \mu_1 < T_n^* \leq \mu_2 \text{ (HS)} \\ T_n &= 2 \text{ if } \mu_2 < T_n^* \leq \mu_3 \text{ (Associate Degree)} \\ T_n &= 3 \text{ if } \mu_3 < T_n^* \leq \mu_4 \text{ (Bachelor's Degree)} \\ T_n &= 4 \text{ if } \mu_4 < T_n^* \leq \mu_5 \text{ (Graduate Degree)} \end{aligned}$$

The μ values represent the thresholds estimated in the analysis, along with the parameter vector β . The thresholds give cutoffs for predicted values of the dependent variable and allow one to predict dependent variable values for specific values of the independent variables.

Elements of the parameter vector β can be interpreted much as slopes in a multiple regression analysis, with positive signs on the parameters indicating a positive relationship between that variable and the dependent variable. Figure 1 shows the relationship between the latent, continuous measure of highest degree T_n^* and the observed highest degree, T_n .

For more information on this model, see Greene (2000).

Independent Variables

Respondent *sex* is measured as Male=1, Female=0.

To measure parental status, we use father's SEI score. In the GSS, this is based on the father's occupation when the respondent was about 16. As noted, the GSS uses computed

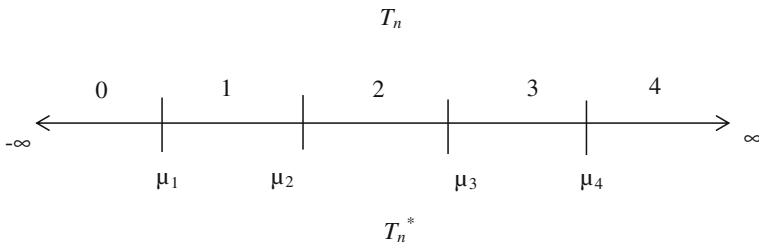


Fig. 1

SEI measures from measures of education, income, and prestige (Blau and Duncan 1967). The NLSY does not use Duncan's SEI, but it does include the 1970 Census Occupational Code, which can readily be converted to the SEI score. The NLSY measures the father's occupational status in 1978, when respondents were 13–21 years of age.

We use three control variables: *age* of the respondent, the respondent's number of *siblings* and the *race* of the respondent (white=1 in GSS, non-Hispanic white=1 in NLSY). *Age* is measured as the respondent's age in years. Given that education increases with age, age is an essential control variable. Since most Americans do not complete their education before they are in their early twenties, we also perform the analyses for only those respondents over 24 years of age. In the NLSY analyses, where possible we analyze educational outcomes for respondents in 1990, by which time the age range of respondents would be 25–33 and respondents likely had completed their education. *Number of siblings* is measured as the respondent's number of brothers and sisters. Previous studies have shown that the number of siblings a person has decreases their level of educational attainment net of other factors (e.g., Steelman and Powell 1989; Downey 1995), so this is an important control variable. This variable also allows us to determine whether parents invest in particular offspring by limiting their total number of offspring.

Results

Tables 1 and 2 give descriptive statistics for all variables used in the analysis from the GSS and the NLSY, respectively (Zero-order correlations between all variables from the GSS are given in the Appendix). Numbers of cases differ for each of the NLSY variables because not all variables were collected for every wave of the survey, and there has been attrition of the sample size over time.

Number of Years of Education

Results of the test of nonbiological investment in children (measured as years of education received) using the GSS data are shown in Tables 3 and 4 and Fig. 2. Table 3 shows the results using years of education as the dependent variable. Model 1 shows significant predictors of years of education are age, race and father's status when the respondent was 16. This model shows that younger respondents, whites and the children of high-status fathers attain more years of education than others. Model 1 also shows a significant interaction between sex and father's status in the positive direction

Table 1 Descriptive statistics for all variables, all cases ($N=11,857$)

Variable	Mean	SD
Years of education	13.58	3.04
Sex (Male=1)	0.45	0.50
Father's SEI	47.10	18.81
Age	47.71	17.28
Race (White=1)	0.81	0.39
Number of siblings	3.56	2.87
<HS	12.99%	
HS	50.69%	
Associate degree	7.81%	
Bachelor's degree	18.47%	
Graduate degree	10.04%	

Source: General Social Survey
2000–2010

predicted by Trivers-Willard. The positive coefficient on the interaction terms indicates that an increase in father's status produces a significantly greater increase in the education of sons than of daughters (Fig. 2).

Model 2 in Table 3 controls for number of siblings. As expected, number of siblings has a significant negative effect on years of education, although the interaction between father's status and respondent sex remains significant. This suggests that parents do not choose to differentially invest in offspring by sex by limiting their total number of offspring. Table 4 gives the same analysis, but only for those over age 24. Once again the positive coefficient on the interaction terms indicates that an increase in father's status produces a significantly greater increase in the education of sons than in the education of daughters. Controlling for number of siblings (Model 2) does not change this result.

Highest Degree Completed

Table 5 shows the results of the analysis using highest degree completed as the measure of nonbiological investment in children and the GSS data. Significant predictors of highest degree attained are age, race, and father's status. This model shows that younger

Table 2 Descriptive statistics for all variables, all cases

Variable	<i>N</i>	Mean	SD
Years of education 1990	5,474	13.11	2.40
Sex (Male=1)	6,110	0.49	0.50
Father's SEI	4,785	40.34	23.90
Age (1979)	6,110	17.60	2.24
Age (1990)	5,488	29.02	2.23
Race (Non-Hispanic white=1)	6,110	0.77	0.42
#Siblings	6,103	3.35	2.29
Public Jr/High school	6,101	0.93	0.25
Tuition 1979	885	1,516.50	1,890.47
Loans 1990	3,161	3,939.70	10,503.17

Source: 1979 National Longitudinal Survey of Youth

Table 3 Unstandardized regression coefficients, regression of number of years of school completed on age, race, sex, parental status, and number of siblings, all cases

Independent variables	Model 1	Model 2
Sex (Male=1)	−0.266† (0.140)	−0.338* (0.135)
Father's SEI	0.054*** (0.002)	0.048*** (0.002)
Age	−0.014*** (0.002)	−0.008*** (0.001)
Race (White=1)	0.665*** (0.067)	0.220** (0.067)
Sex × father's SEI	0.006* (0.003)	0.007* (0.003)
Number of siblings		−0.256*** (0.009)
R^2	0.153	0.205
N	11,843	11,843

Standard errors in parentheses

Source: General Social Survey 2000–2010

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

individuals, whites, and the children of high-status fathers attain higher degrees than others. In this model, the interaction between sex and father's status is not significant. Model 2 adds number of siblings as a control variable. Once again, number of siblings is negatively associated with highest degree attained. In this model the interaction of sex by father's status is significant, showing that the sons of high-status fathers attain higher degrees than the daughters of high-status fathers.

Table 6 shows the same analysis for those over 24 years of age. In both Model 1 and Model 2 the interaction of sex by father's status is significant, showing that sons of high-status fathers attain higher degrees than daughters. Figure 3 illustrates this relationship, showing the percentage of sons and daughters of the highest- and lowest-status fathers who attain college degrees and beyond.

Table 4 Unstandardized regression coefficients, regression of numbers of years of school completed on age, race, sex, parental status, and number of siblings, cases age >24

Independent variables	Model 1	Model 2
Sex (Male=1)	−0.277† (0.148)	−0.348* (0.143)
Father's SEI	0.056*** (0.002)	0.048*** (0.002)
Age	−0.025*** (0.002)	−0.019*** (0.002)
Race (White=1)	0.718*** (0.071)	0.245** (0.071)
Sex × father's SEI	0.006* (0.003)	0.007** (0.003)
Number of siblings		−0.263*** (0.010)
R^2	0.168	0.222
N	10,857	10,857

Standard errors in parentheses

Source: General Social Survey 2000–2010

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

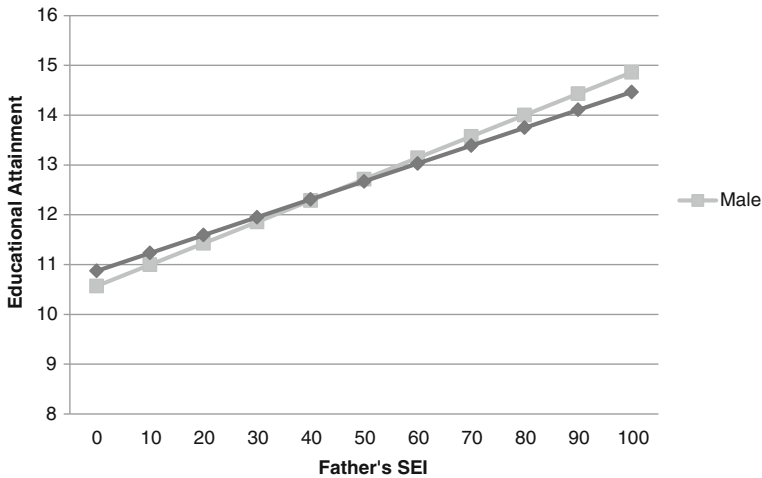


Fig. 2 Educational attainment by father's SEI. 2000–2010 GSS. Years of education predicted using Model 1, Table 6, using means for all variables

Type of High School, College Tuition, and College Loans

To understand the mechanism by which sons of high-status fathers receive more education than daughters, we further analyze the effects of father's status on whether the respondent attended public or private high school, the amount of tuition paid at the respondent's last college, and the amount of total educational loans (in 1990) using data from the NLSY. Table 7 shows the results of this analysis.

Table 5 Ordered probit parameter estimates, regression of highest degree completed on age, race, sex, parental status, and number of siblings, all cases

Independent variables/Parameters	Model 1	Model 2
Threshold 1 (High school)	−0.187 (0.050)	−0.697 (0.054)
Threshold 2 (Associates/Junior college)	1.418 (0.051)	0.967 (0.054)
Threshold 3 (Bachelor's)	1.659 (0.051)	1.213 (0.054)
Threshold 4 (Graduate)	2.437 (0.053)	2.009 (0.056)
Sex (Male=1)	−0.090 (0.054)	−0.116* (0.055)
Father's SEI	0.020*** (0.001)	0.019*** (0.001)
Age	−0.002*** (0.001)	−0.0002 (0.001)
Race (White=1)	0.225*** (0.026)	0.068* (0.027)
Sex × father's SEI	0.002† (0.001)	0.002* (0.001)
Number of siblings		−0.096*** (0.004)
Pseudo R^2	0.054	0.074
N	11,857	11,857

Standard errors in parentheses

Source: General Social Survey 2000–2010

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

Table 6 Ordered probit parameter estimates, regression of highest degree completed on age, race, sex, parental status, and number of siblings, cases age >24

Independent variables/Parameters	Model 1	Model 2
Threshold 1 high school	−0.461 (0.054)	−1.030 (0.059)
Threshold 2 Associates/Junior college	1.129 (0.055)	0.627 (0.058)
Threshold 3 bachelor	1.374 (0.055)	0.879 (0.058)
Threshold 4 graduate	2.158 (0.057)	1.682 (0.060)
Sex (Male=1)	−0.099† (0.057)	−0.126* (0.057)
Father's SEI	0.021*** (0.0001)	0.019*** (0.001)
Age	−0.008*** (0.001)	−0.006*** (0.001)
Race (White=1)	0.241*** (0.027)	0.072* (0.028)
Sex × father's SEI	0.002* (0.001)	0.003* (0.001)
Number of siblings		−0.100*** (0.004)
Pseudo R^2	0.062	0.084
N	10,871	10,871

Standard errors in parentheses

Source: General Social Survey 2000–2010

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

Model 1 in Table 7 replicates the findings reported previously from the GSS data. The effect of father's status on years of education is positive for both sons and daughters, but the significant interaction between father's status and respondent sex shows that the sons of high-status fathers attain more years of education than the daughters, and conversely, daughters of low-status fathers attain more education than the sons.

Model 2 uses the NLSY data on whether the respondent's current or last junior high or high school was public or private. The main effect of father's status is not significant, showing that respondents with high-status fathers are not more or less likely to attend

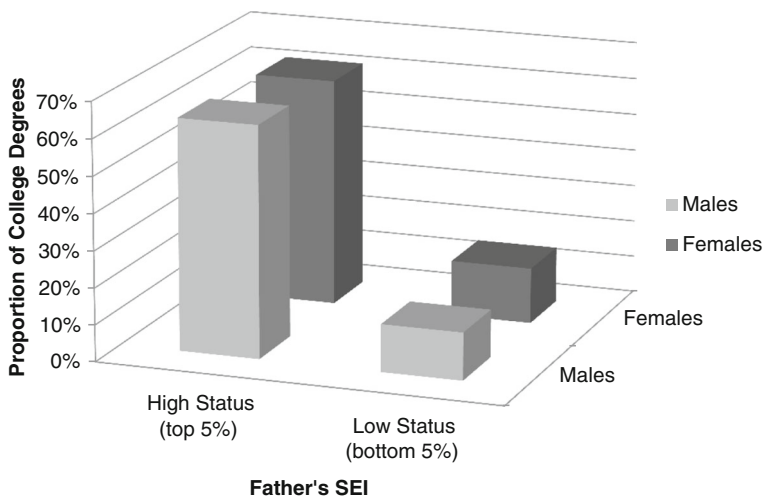
**Fig. 3** Proportion of college degrees by sex. 2000–2010 GSS. All cases age >24

Table 7 Unstandardized regression coefficients, regression of number of years of school completed, type of schooling, amount of tuition and amount of loans on independent variables

Independent variables	Dependent variables			
	Years of education (1990)	Type of Jr/High school ^a	Amount of tuition (1979)	Amount of loans (1990)
	Model 1	Model 2	Model 3	Model 4
Age	0.053*** (0.015)	0.015 (0.028)	10.693 (27.003)	−285.370* (110.294)
Sex (Male=1)	−0.305* (0.126)	−0.017 (0.240)	−68.372 (235.306)	−1484.892 (968.902)
Father's status	0.036*** (0.002)	0.000 (0.004)	0.394 (3.581)	−3.282 (14.733)
Sex × father's status	0.007* (0.003)	−0.0004 (0.005)	2.418 (4.727)	31.906† (20.029)
Race (White=1)	−0.025 (0.089)	0.091 (0.166)	249.312 (159.935)	−404.553 (664.301)
Number of siblings	−0.174*** (0.016)	0.030 (0.031)	−41.210 (28.085)	−38.842 (117.773)
R ²	0.210	0.001	0.012	0.005
N	4,292	4,297	686	2,236

Standard errors in parentheses

Source: 1979 National Longitudinal Survey of Youth

† $p < 0.1$ * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$

^a (public=1, private=0)

public school. The interaction effect between sex and father's status is also not significant, showing that the effect of parental status on attending public school does not differ by sex of respondent.

Model 3 shows the analysis of college tuition. This sample size is small because the NLSY only asked this question of respondents in 1979, at which time few of the respondents were in college. Neither father's status or the interaction of sex and father's status have significant effects on the amount of college tuition paid.

Model 4 shows the analysis of the respondent's loan amount in 1990. The only significant predictor of loan amount is the respondent's age. The interaction of father's status with respondent sex approaches significance ($p=0.111$), and the direction of the coefficient suggests that the sons of high-status fathers have higher loan amounts than daughters.

In sum, these analysis suggest that neither the type of high school attended, the college tuition paid, nor the amount of loans the respondent had in 1990 are part of the mechanism by which the sons of high-status fathers attain more education than the daughters while the daughters of low-status fathers attain more education than the sons.

Biological Investment

As discussed previously, if high-status fathers are more likely to have sons among their biological offspring, then in a random sample of the population males will be more likely than females to have high-status fathers. Table 8 shows the correlation between father's status and respondent sex. As predicted, there is a significant, positive correlation between sex and father's status ($r=0.037$). These are similar to correlations previously found between father's status and respondent sex (Hopcroft 2005). For the earlier years of the GSS, they were $r=0.031$ and $r=0.034$. Although the correlation is

weak, it is consistently significant across the 40 years of the GSS and suggests that men are more likely to have high-status fathers than women.

Discussion and Conclusion

This analysis finds that in the contemporary U.S., the sons of high-status fathers obtain more education and higher degrees than the daughters of high-status fathers, and conversely, the daughters of low-status fathers obtain more education and higher degrees than the sons of low-status fathers. These findings for education are similar to Cox (2003) and Hopcroft (2005) for earlier years (1972–2000) of the General Social Survey. Given the context of the contemporary United States, with its highly diverse education market and the role of parents in funding the education of their children, we suggest that these results reflect level of family investment and are consistent with the prediction of Trivers-Willard that high-status families will invest more in boys, and low-status families will invest more in girls.

It is difficult to think of an alternative explanation for these findings. Given traditional sex roles, which tend to be more pronounced in the lower and working classes, it might be predicted that the sons of lower-class men would obtain more education than the daughters. Further, given that elite families are least likely to subscribe to traditional sex roles and have historically been more likely to educate daughters than lower-class families, it might be predicted that the sons and daughters of elite families would receive the same amount of education. The fact that women are now the majority of college students (Goldin et al. 2006) may help explain why the daughters of lower-status men obtain more education than sons, but it cannot help explain why the sons of higher-status men obtain more education than the daughters. Rather, this seems to be a true Trivers-Willard effect of differential investment in offspring by sex.

The differences in amount of education received by sons and daughters at the upper and lower end of socioeconomic scale are small, but we argue that these results likely underestimate the true differences. This is because our measures of education attained do not measure the qualitative value of the education received (quality of years of education or quality of degree). Thus, the measures of education used equate 2 years at a prestigious Ivy League college with 2 years of community college, yet they likely entail differential amounts of parental investment. It may be that the sons of high-status fathers also receive a higher-quality education than the daughters of high-status fathers, and vice versa for offspring of low-status fathers, but there is no way to determine that from this analysis. Future waves of the GSS will include the name of the college attended by respondents, so future analysts will be able to examine this issue. Unfortunately, we do not have that information in the 2000–2010 GSS or in the 1979 NLSY.

One strength of the measure of parental investment used here (education attained) is that it is an outcome measure that captures a large part of parental investment.

Table 8 Correlation between respondent sex and occupational status of father

	Father's SEI
Sex (Male=1)	0.037***
N	11,926

*** $p < 0.001$

However, we were unable to determine the specific mechanism by which the results are achieved. Neither type of high school, amount of tuition paid for 1 year of college, nor total amount of college loans account for the differences in the amount of education attained by sons and daughters. Also, the data set we used does not allow us to determine if there is any sex difference in educational investment in children within the same family. Future studies should obtain more detailed information on respondents' and siblings' education and find out whether high-status families are more likely to encourage and/or support additional education or higher-quality education for sons than for daughters and low-status families are more likely to encourage and/or support additional education or higher-quality education for daughters rather than sons.

This analysis also finds that in the contemporary U.S., high-status men are more likely to have sons than daughters. This finding also supports the Trivers-Willard hypothesis. The sex ratio findings are consistent with studies that have found more boys born to affluent mothers and the wives of billionaires (Almond and Edlund 2007; Cameron et al. 2009). Hopcroft (2005) also found more boys born to high-status men.

Once again, the mechanism by which this result comes about is unclear. It may be that high-status men experience less stress and therefore are more likely to father boys, perhaps because of changes in sperm motility (Fukuda et al. 1998) or changes in the frequency and timing of coitus (Lazarus 2002). Alternatively, the female partners of high-SES men may be less likely to experience stress and are thus more likely to conceive a male and/or carry a male fetus to term. There is evidence that maternal stress lowers the sex ratio, resulting in more girls being born (Catalano 2003; Catalano and Bruckner 2005; Hansen et al. 1999; Khashan et al. 2009; Subbaraman et al. 2010; Torche and Kleinhaus 2012).

The findings presented here suggest that evolutionary hypotheses such as the Trivers-Willard hypothesis are relevant to contemporary populations such as the U.S. We suggest that more fine-grained measures of education attained—public versus private colleges attended, selectivity of college, etc.—would reveal further Trivers-Willard effects in the U.S.

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Appendix

Table 9 Correlations, all variables (GSS)

Variable	Years of education	Sex (Male=1)	Father's status	Age	Race (White=1)	Number of siblings
Years of education	1					
Sex (Male=1)	0.02*	1				
Father's status	0.38***	0.04***	1			
Age	−0.12***	−0.04***	−0.13***	1		
Race (White=1)	0.12***	0.02†	0.13***	0.13***	1	
Number of siblings	−0.32***	−0.04***	−0.22***	0.13***	−0.24***	1

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